Class 10 Math - Chapter 10: Circles

🌟 Introduction

A **circle** is a collection of all points strictly in a plane that are at a constant exact distance (radius) from a fixed constant point (center).

In Class 10, the focus shifts sharply to the relationship between a sweeping circle and a perfectly straight line intersecting it.

Line and Circle Relationships:

Given a circle and a line in a plane, three absolute possibilities exist:

1. **Non-intersecting Line:** The line never touches the circle.
2. **Secant:** The line dynamically intersects the circle at exactly TWO distinct points.
3. **Tangent:** The line gracefully touches the circle at exactly ONE single point.

📌 Tangent to a Circle

A tangent to a circle is a special line that intersects the circle at exactly one single point, known as the **point of contact**.

> [!NOTE] There is exactly **only one** tangent at any given single point on a specific circle. A circle therefore has an infinite mathematical number of tangents, mapping to its infinite points!

Mind Map of Tangents

```
graph TD;
    Point_Location-->Inside["Inside Circle"];
    Point_Location-->On_Boundary["On Boundary"];
    Point_Location-->Outside["Outside Circle"];
    Inside-->Zero_Tangents["0 Tangents\n(Lines become secants)"];
    On_Boundary-->One_Tangent["1 Tangent"];
    Outside-->Two_Tangents["2 Tangents\n(Equal in length)"];
```

⚡ Key Theorems of Circles

These two theorems form the absolute logical backbone of almost every single proof and numerical problem in this chapter:

Theorem 10.1: The Radius-Tangent Theorem

> "The tangent at any exact point of a circle is structurally perpendicular to the radius acting strictly through the point of contact."

Mathematical Implication: If XY is a tangent at point P, and O is the center of the circle, then $OP \perp XY$. This allows you to construct Right-Angled triangles and use precisely the Pythagorean theorem in circle problems!

Theorem 10.2: The External Point Theorem

> "The absolute lengths of tangets drawn strictly from an external point directly to a circle are perfectly equal."

Mathematical Implication: If structurally tangents PA and PB are drawn from an outside point P to a precise circle with center O, then $PA = PB$.

> [!IMPORTANT] The triangles $\triangle OAP$ and $\triangle OBP$ formed by these tangents and the radii are dynamically **congruent** by the RHS congruence rule. As a result, $\angle OPA = \angle OPB$ (the center lies precisely on the angle bisector!).

❌ Common Mistakes

> [!WARNING]
>
> 1. **Pythagoras Misapplication:** Because $OP \perp XY$, the hypotenuse is uniquely the line connecting the *center* to the *external point*. Students occasionally treat the tangent line length itself as the hypotenuse!
> 2. **Forgetting Congruence:** Many complex geometry proofs require you to remember that since $PA = PB$, the resulting triangle $\triangle PAB$ is universally **isosceles**.
> 3. **Concentric Circles:** When dealing logically with two concentric circles, the chord of the larger circle that directly touches the smaller circle acts geometrically as a *tangent* to the smaller circle.

📝 Quick Revision

Concept	Property
Number of Tangents from Inside	0
Number of Tangents from Surface	1
Number of Tangents from Outside	2 (Both are strictly equal in length)
Angle between Tangent and Radius	Exactly 90°

Summary

The magic of circle geometry in Class 10 primarily links straight lines securely to curved surfaces through exactly one point. By remembering that a tangent always creates a perfectly crisp 90° angle with its host radius, you unlock the powerful ability to deploy standard right-triangle trigonometry and Pythagoras rules directly into curved environments!